

Assignment 2

Course Name: COM 837, ML for Wireless Communication

Deadline: 11th September, EOD

Question 1 - SVM for LOS/NLOS Identification

Reference Paper: C. Huang, A. F. Molisch, R. He, R. Wang, P. Tang, B. Ai, and Z. Zhong, “Machine Learning-Enabled LOS/NLOS Identification for MIMO Systems in Dynamic Environments,” *IEEE Transactions on Wireless Communications*, vol. 19, no. 6, pp. 3643–3657, Jun. 2020.

Dataset columns:

Column	Description
snr_db	SNR in dB. Filter on this column to work at a specific SNR.
label	+1 for LOS, –1 for NLOS. This is your SVM target variable.
h_real_0 ... h_real_5	Real part of each noisy tap amplitude.
h_imag_0 ... h_imag_5	Imaginary part of each noisy tap amplitude.
tau_0 ... tau_5	Tap delays in nanoseconds.

[Dataset Generation Program](#)

Part (a):

You must extract five scalar features from each channel realisation in the dataset.

- Kurtosis of the received power
- Skewness of the received power
- Rising time
- Root mean square-delay spread
- Rician K-factor

Add these columns to the dataframe. All subsequent questions use these features.

Part (b):

This question asks you to understand how much each feature contributes to classification on its own, and what is gained by combining them.

1. Train 6 SVM classifiers, all using an RBF kernel with $C = 1$:

- **SVM-1:** trained on Kurtosis only
- **SVM-2:** trained on Skewness only
- **SVM-3:** trained on Rising time only
- **SVM-4:** trained on Root mean square-delay spread only
- **SVM-5:** trained on Rician K-factor only
- **SVM-6:** trained on all five features combined

For each SNR value in the dataset, train on that SNR's data and test on a held-out split (80/20 train-test split). Always apply **StandardScaler** before training. Fit the scaler on the training set and apply it to the test set.

2. Plot Classification Accuracy (%) vs. SNR (dB) for all six classifiers on a single figure.
3. Answer the following:
 - Which single feature performs best overall? Why does that feature separate LOS from NLOS better than the others, given the channel model used?
 - Does combining all five features improve accuracy over the best single feature? At which SNR values is the improvement most noticeable?
 - One of the five features is likely to perform poorly at low SNR even though it is physically meaningful. Which one, and why?

Part (c):

1. Using the combined five-feature SVM from Part (b) (SVM-6), train once using only data from SNR = 25 dB. Then evaluate this fixed model on the test sets at every SNR value from 0 to 30 dB.
2. On the same figure, plot two curves:
 - **“Train = Test SNR”** - accuracy at each SNR when the model is trained and tested at the same SNR (reuse Part (b) results for SVM-6)
 - **“Train at 25 dB”** - accuracy at each SNR when the model is only trained at 25 dB
3. Answer the following:
 - At high SNR, do the two curves agree? What does this tell you about the features at high SNR?
 - At low SNR, which strategy performs better and by how much? Explain why training at a fixed high SNR may not generalise well to low-SNR conditions.
 - Would training at a low SNR (e.g., 0 dB) and testing across all SNR give a different result? Reason through this without necessarily running the experiment.

Question 2 - Constellation Demodulation & Clustering using K-means

Dataset Generation

You must write a Python script to synthesize complex baseband 16-QAM signal realizations over an Additive White Gaussian Noise (AWGN) channel. Generate a standard square 16-QAM constellation grid with coordinates drawn from $\{-3, -1, 1, 3\}$. Scale the complex points so the average symbol energy is 1. For SNR values in $[0, 5, 10, 15, 20, 25, 30]$ dB, add zero-mean complex AWGN to the transmitted symbols. Generate 200 samples per constellation point for every SNR value. Set the random seed to 67.

Part (a):

You must extract two scalar features from each received noisy sample in the dataset.

- Instantaneous Amplitude (r)

$$r = \sqrt{Rx_I^2 + Rx_Q^2} \quad (1)$$

- Instantaneous Phase (θ) in radians

$$\theta = \arctan\left(\frac{Rx_Q}{Rx_I}\right) \quad (2)$$

Add these columns to the dataframe. All subsequent questions use these features along with the Cartesian features.

Part (b):

This question asks you to evaluate how effectively K-means identifies cluster partitions and whether Cartesian or Polar representations best capture digital modulation geometries.

1. Isolate the data subset corresponding to SNR = 25 dB. Fit standard K-means models using Cartesian coordinates for cluster counts ranging from $K = 2$ to $K = 20$ (use `k-means++` initialization with `n_init=10`, `random_state=42`).
2. On a single figure with two side-by-side subplots, plot:
 - Within-Cluster Sum of Squares (Inertia) vs. K
 - Silhouette Coefficient vs. K
3. Set $K = 16$. Fit the K-means algorithm on the Cartesian space at SNR = 25 dB. Generate a 2D scatter plot of the received samples colored by their assigned cluster index. Overlay the 16 learned cluster centroids as distinct markers.
4. Answer the following:
 - Compute the **Cluster Purity** comparing the K-means cluster predictions to the ground-truth `symbol_id` using three distinct feature sets: Feature Set 1 (`rx_I`, `rx_Q`), Feature Set 2 (r , θ), and Feature Set 3 (`rx_I`, `rx_Q`, r , θ). (*Hint: To calculate purity, assign each cluster to the ground-truth symbol most frequent within it, then compute the overall fraction of correctly assigned samples*). Apply `StandardScaler` to all features in Set 3 before clustering. Which feature set produces the highest Cluster Purity?
 - Why does standard Euclidean distance on Cartesian coordinates naturally fit QAM grids, whereas standard Euclidean distance on unweighted Polar coordinates can lead to distorted decision boundaries?

Part (c):

1. Using $K=16$, evaluate two demodulation strategies.
 - **Strategy 1:** For each SNR value in the dataset (0 to 30 dB), fit a fresh K-means model directly on that SNR's Cartesian data and compute its Cluster Purity against `symbol_id`.
 - **Strategy 2:** Fit the K-means model once using only the dataset at SNR = 25 dB. Use these fixed centroids to predict the cluster labels for every individual SNR value (0 to 30 dB) and compute the Cluster Purity.
2. On the same figure, plot two curves:
 - **"Adaptive K-Means"** - Cluster Purity at each SNR when a fresh model is fitted per SNR
 - **"Fixed Template"** - Cluster Purity at each SNR when using the fixed centroids from 25 dB
3. Answer the following:
 - At high SNR, do the two curves agree? What does this indicate about centroid stability in low-noise regimes?
 - At low SNR, which strategy achieves higher classification accuracy?
 - Explain geometrically why fitting an unsupervised clustering algorithm directly on low-SNR samples causes centroid merging and high classification error compared to using a fixed geometric template.

Academic Integrity

Plagiarism and the use of any Generative AI tool (ChatGPT, Copilot, Gemini, Claude, etc.) to write or generate code or report text is strictly prohibited. Individual viva-voce sessions will be conducted for every student. You will be expected to explain any line of code and every observation in your report. **Inability to do so will result in zero marks for the entire assignment.**

1. **Do not modify `generate_dataset.py`.** Changing the channel parameters or the random seed produces a different dataset and will invalidate comparisons during the viva.
2. **Code quality is graded.** All code must be well-commented with self-explanatory variable names. Uncommented or obfuscated code receives zero marks for that question regardless of numerical correctness.
3. **Observations are mandatory for every question.** Each figure must be accompanied by a written discussion in the PDF report. Submitting code and plots *without* written analysis receives zero marks for the observation sub-parts.
4. **What to submit on the course portal:**
 - The URL of your GitHub repository (must be **public**).
 - A single PDF report containing all figures and written observations, clearly numbered by question.